allows them to develop industry-specific algorithms in sectors such as "banking, manufacturing, logistics, life sciences, and energy".

TCS, meanwhile, has been at this game for a while. Sharma notes that the company's quantum credentials go back to early collaborations with DRDO and TIFR, and now with IBM, AWS, and others. The goal? "To grow India's quantum capacity – from labs to live enterprise pilots," Sharma says.

These aren't moonshot pet projects. LTIMindtree has already conducted pilot projects with global clients across sectors like banking, energy, and pharmaceuticals. That's real progress.

Quantum Computing in India: Key developments so far

- The National Quantum Mission, launched in 2023 with a budget of ₹6,003 crore, targets the development of intermediate-scale quantum computers (50-1,000 qubits) by 2031.
- In April 2025, Bengaluru-based startup QpiAI launched QpiAI-Indus, India's most powerful quantum computer with a 25-qubit superconducting processor—a key step in indigenous development.
- Quantum technologies in India are being applied to fields like secure communications, drug discovery, material science, finance, and logistics, aiming to accelerate solutions across key sectors.
- At least eight leading Indian startups, including QNu Labs and QpiAI, are pioneering advances in quantum computing, communication, and hardware, often with government backing and patents filed.
- Institutions like TIFR Mumbai, IITs, and the Raman Research Institute are core to India's quantum research.

Quantum-safe by design

As quantum computers inch closer to breaking today's encryption, both TCS and LTIMindtree are preparing for the so-called Y2Q moment – the day current cryptography fails.

"Much like Y2K catalyzed a massive shift in global technology preparedness, the coming 'Y2Q' moment... calls for immediate attention," warns Sharma. "We have built advanced quantum risk management capabilities and forged strategic partnerships within our Co-Innovation Network (COIN) to stay ahead in this evolving field".

At TCS, this means a full-stack approach: "structured pilot programs that assess existing cryptographic infrastructure, identify vulnerabilities, and implement post-quantum algorithms... often in collaboration with leading ecosystem partners," Sharma adds. It's not just about solving math problems; it's about fortifying national security and enterprise resilience in time.

LTIMindtree's approach is equally structured – almost methodical. "Our development cycle for quantum-safe cryptography is structured around four key phases: industry-specific use-cases, research & prototyping, standards alignment, and deployment," explains Mitra.

They prioritize high-stakes industries like financial services, healthcare, and critical infrastructure, where data longevity makes them prime targets for "harvest now, decrypt later" attacks. Their internal teams prototype solutions with partners like PQShield and Quantum Xchange, aligning with evolving NIST and IETF guidelines.

Homegrown talent, global context

India's biggest constraint in quantum computing isn't infrastructure. It's talent. That's why both firms are pouring resources into skilling.

"We were one of the first in the industry to recruit from the first graduating batch of the M.Tech. in Quantum Technology at IISc Bangalore," says Mitra. LTIMindtree's hiring strategy leans on internships, academic partnerships, and in-house upskilling to create a "quantum-ready workforce".

TCS is thinking similarly but at scale. "We are investing in strategic alliances and national skilling programs," says Sharma, noting that ecosystem-building is as crucial as algorithm development.

India, quantum-first

If the first internet wave in India was about leapfrogging fixed-line infrastructure into mobile-first access, this quantum wave is about reimagining India not just as a user, but as a maker.

"We are helping shape an ecosystem where India moves from being quantum innovation consumer to becoming a global contributor," Sharma says.

It's a tall order, but if the last three decades of India's tech evolution have taught us anything, it's this – give a billion minds the tools and time, and they'll crack more than just code... they'll rewrite the rules. In the quantum future, India isn't waiting to be invited to the table. It's busy building its own. d